torch.from_file#

https://pytorch.org/docs/stable/generated/torch.from_file.html

Description:

Creates a CPU tensor with a storage backed by a memory-mapped file. If shared is True, then memory is shared between processes. All changes are written to the file. If shared is False, then changes to the tensor do not affect the file. size is the number of elements in the Tensor. If shared is False, then the file must contain at least size * sizeof(dtype) bytes. If shared is True the file will be created if needed. Only CPU tensors can be mapped to files.

Function Signature

```
torch.from_file(filename, shared=None, size=0, *,
dtype=None, layout=None, device=None, pin_memory=False)#
Parameters
Keyword Arguments
```

Parameters

Keyword

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. pin_memory (bool, optional) – If set, returned tensor would be allocated in the pinned memory. Works only for CPU tensors. Default: False.

Code Examples

```
>>> t = torch.randn(2, 5, dtype=torch.float64)
>>> t.numpy().tofile('storage.pt')
>>> t_mapped = torch.from_file('storage.pt', shared=False,
size=10, dtype=torch.float64)
```

Notes & Warnings

NOTE Note Only CPU tensors can be mapped to files.

NOTE Note For now, tensors with storages backed by a memory-mapped file cannot be created in pinned memory.

Detailed Documentation

Documentation

Creates a CPU tensor with a storage backed by a memory-mapped file.

If shared is True, then memory is shared between processes. All changes are written to the file. If shared is False, then changes to the tensor do not affect the file.

size is the number of elements in the Tensor. If shared is False, then the file must contain at least size * sizeof(dtype) bytes. If shared is True the file will be created if needed.

Only CPU tensors can be mapped to files.

For now, tensors with storages backed by a memory-mapped file cannot be created in pinned memory.

filename (str) – file name to map

shared (bool) – whether to share memory (whether MAP_SHARED or MAP_PRIVATE is passed to the underlying mmap(2) call)

size (int) – number of elements in the tensor

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

pin_memory (bool, optional) – If set, returned tensor would be allocated in the pinned memory. Works only for CPU tensors. Default: False.